

Q1)  $X_1 \dots X_n \sim \text{Gamma}(\alpha, \beta)$   $\alpha, \beta > 0$  A46

$$f(x|\alpha, \beta) = \frac{1}{\Gamma(\alpha)\beta^\alpha} x^{\alpha-1} e^{-x/\beta} \quad \mathbb{P}_{(0, \infty)}(x) = \frac{1}{\Gamma(\alpha)\beta^\alpha} \frac{1}{x} \exp\left\{-\frac{x}{\beta} + \alpha \log(x)\right\}$$

a)  $a(\theta) = \frac{1}{\Gamma(\alpha)\beta^\alpha}$

$$b(x) = \frac{1}{x}$$

$$b_1 = \frac{1}{\beta}$$

$$\tau_1 = x$$

$$b_2 = 2$$

$$\tau_2 = \log(x)$$

b)  $T = \left( \sum_{i=1}^n \tau_1(x_i), \sum_{i=1}^n \tau_2(x_i) \right)$   
 $= \left( \sum_{i=1}^n x_i, \sum_{i=1}^n \log(x_i) \right)$   
 $= \left( \sum_{i=1}^n x_i, \log\left(\prod_{i=1}^n x_i\right) \right)$

c)  $\mathbb{E}(x_i) \rightarrow \mathbb{E}\left(\frac{\partial}{\partial \beta} \{b_1 \tau_1 + b_2 \tau_2\}\right) = -\frac{\partial}{\partial \beta} \log(a(\theta))$

$$\mathbb{E}\left(\frac{x_i}{\beta^2}\right) = +\frac{\partial}{\partial \beta} \left\{ \log(\Gamma(\alpha)) + \alpha \log(\beta) \right\}$$

$$\mathbb{E}(x_i) = \beta^2 \cdot \frac{\alpha}{\beta} = \alpha \beta$$

$$\rightarrow \mathbb{E}(\sum x_i) = \sum \mathbb{E}(x_i) = \sum \alpha \beta = n \alpha \beta$$

$$Var(x_i) \Rightarrow Var\left(\frac{x_i}{\beta^2}\right) = +\frac{\partial^2}{\partial \beta^2} \left\{ \log(\Gamma(\alpha)) + \alpha \log(\beta) \right\} - \mathbb{E}\left(\frac{\partial}{\partial \beta} \frac{x_i}{\beta^2}\right)$$

$$\frac{1}{\beta^4} Var(x_i) = -\frac{2}{\beta^2} + \frac{2}{\beta^3} \mathbb{E}(x_i)$$

$$Var(x_i) = \beta^4 \left\{ -\frac{2}{\beta^2} + \frac{2\alpha}{\beta^2} \right\} = \beta^4 \left\{ \frac{2}{\beta^2} \right\} = 2\beta^2$$

$$Var(\sum x_i) = \sum Var(x_i) = \sum 2\beta^2 = 2n\beta^2$$

d) In 2 Concord 
$$\bar{I}_X(\theta) = \frac{1}{n} \left\{ \frac{\partial^2}{\partial \theta^2} \log(f(x|\theta)) \right\} = E \left[ \left( \frac{\partial}{\partial \theta} \log f(x|\theta) \right)^2 \right]$$

$n I_{X_1}(\theta)$

$$\log f(x|\theta) = -\log(\Gamma(2)) - 2 \log(\theta) + (2-1) \log(x) - \frac{x}{\theta}$$

$$\frac{\partial}{\partial \theta} \log f(x|\theta) = -\frac{2}{\theta} + \frac{x}{\theta^2} = \frac{x-2\theta}{\theta^2}$$

$$\frac{\partial^2}{\partial \theta^2} \log f(x|\theta) = +\frac{2}{\theta^2} - \frac{2x}{\theta^3} = \frac{2\theta - 2x}{\theta^3}$$

$$-E \left( \frac{\partial^2}{\partial \theta^2} \log f(x|\theta) \right) = -\frac{2}{\theta^2} + \frac{2}{\theta^3} E(x) = -\frac{2}{\theta^2} + \frac{2}{\theta^3} 2\theta = \frac{2(2-2)}{\theta^2} = \frac{2}{\theta^2}$$

$$I_X(\theta) = \frac{2}{\theta^2}$$

$$I_X(\theta) = \text{Var}(\log f(x|\theta)) \quad I_{11} = E \left[ \left( \frac{\partial}{\partial \theta} \log f(x|\theta) \right)^2 \right]$$

e) For independent  $\alpha, \beta$

$$\log f(x|\alpha, \beta) = -\log(\Gamma(2)) - 2 \log(\beta) + (2-1) \log(x) - \frac{x}{\beta}$$

$$\frac{\partial \log f}{\partial \beta} = \frac{(x-2\beta)}{\beta^2} \quad \frac{\partial \log f}{\partial \alpha} = -\frac{\Gamma'(2)}{\Gamma(2)} - \log(\beta) + \log(x)$$

$$\frac{\partial^2 \log f}{\partial \beta \partial \alpha} = -\frac{1}{\beta^2} = -\frac{1}{\beta} \quad \frac{\partial^2 \log f}{\partial \alpha^2} = -\frac{1}{\beta} \quad \frac{\partial^2 \log f}{\partial \beta^2} = \frac{2\beta - 2x}{\beta^3}$$

$$I_{11} = \psi'(2)$$

$$I_{21} = -1/\beta$$

$$I_{22} = 2/\beta^2$$

$$\bar{I}_{11}$$

$$I_{\alpha\beta} = \begin{bmatrix} \psi'(2) & -1/\beta \\ -1/\beta & 2/\beta^2 \end{bmatrix}$$

$$\frac{\partial^2 \log f}{\partial \alpha^2} = \psi'(2)$$

$$I_X(\alpha, \beta) = \begin{bmatrix} \psi'(2) & -1/\beta \\ -1/\beta & 2/\beta^2 \end{bmatrix}$$

5) 2 Casos:  $T = \sum_{i=1}^n X_i$   $X_i$  iid de orde

$$Z = \frac{X}{B} \rightarrow BZ = X \quad \frac{dX}{dz} = B$$

$$f_Z(z) = f_X(A^{-1}(z)) \frac{dA^{-1}}{dz} = \frac{1}{\Gamma(\alpha) B^\alpha} (Bz)^{\alpha-1} \exp\left\{-\frac{Bz}{B}\right\} \cdot \frac{1}{B}$$

$$= \frac{1}{\Gamma(\alpha)} z^{\alpha-1} \exp\{-z\} \quad \text{no depende de } B$$

$\frac{X_{(1)}}{B} \rightarrow$  no depende de  $B$ .

$\rightarrow \frac{T}{B} = \sum \frac{X_i}{B}$  no depende de  $B$

$\rightarrow \left(\frac{X_{(1)}}{T}\right) = \frac{X_{(1)}/B}{T/B} \xrightarrow{\text{Dado}} \text{no depende de } B, T, \text{ completo}$

$$E\left(\frac{X_{(1)}}{T}\right) = E\left(\frac{X_{(1)}}{T} \cdot T \mid T\right) = E\left(\frac{X_{(1)}}{T} \mid T\right) \cdot E(T \mid T)$$

$$= E\left(\frac{X_{(1)}}{T} \mid T\right) \cdot T$$

$$= E\left(\frac{X_{(1)}}{T}\right) \cdot T \stackrel{*}{=} T \cdot \frac{E(X_{(1)})}{E(T)}$$

(\*)  $\therefore \frac{X}{Y} \perp Y$

$$E(X^k) = E\left(\frac{X^k}{Y^k} \cdot Y^k\right) = E\left(\frac{X^k}{Y^k}\right) \cdot E(Y^k)$$

$$\rightarrow \frac{E(X^k)}{E(Y^k)} = E\left(\frac{X^k}{Y^k}\right)$$

$$P2) \quad X = \{1, 2, 3\}^{100}$$

$$\theta \in \Theta = [0, 1]$$

$$P(X=1) = \theta^2$$

$$P(X=2) = 2\theta(1-\theta)$$

$$P(X=3) = (1-\theta)^2$$

$$X=1 \Rightarrow \text{Mucche curi}$$

$$X=2 \Rightarrow \text{pece curi}$$

$$X=3 \Rightarrow \text{sin pasta}$$

Ident, parallel

$$\sum_{X, N \in \mathcal{X}}$$

$$P_r(\vec{X}|\theta) = P(\vec{X}|\theta) \rightarrow$$

$$\theta_1, \theta_2 \in \Theta$$

2)

$$P_{\vec{X}}(\vec{X}|\theta) = \theta^{2 \cdot N_1} (2\theta(1-\theta))^{N_2} (1-\theta)^{2N_3}, \quad N_1 = \sum_{i: X_i=1} 1$$

$$P(\vec{X}|\theta_1) = P(\vec{X}|\theta_2)$$

$$\theta_1^{2N_1} (2\theta_1(1-\theta_1))^{N_2} (1-\theta_1)^{2N_3} = \theta_2^{2N_1} (2\theta_2(1-\theta_2))^{N_2} (1-\theta_2)^{2N_3}$$

$$\theta_1^{2N_1} (2\theta_1(1-\theta_1))^{N_2} (1-\theta_1)^{2N_3} = \theta_2^{2N_1} (2\theta_2(1-\theta_2))^{N_2} (1-\theta_2)^{2N_3}$$

$$N_2 = \sum_{i: X_i=2} 1$$

$$N_3 = 100 - N_1 - N_2$$

$$\left(1 - \frac{\theta_1 - \theta_2}{1 - \theta_2}\right)$$

$$\left(\frac{\theta_1}{\theta_2}\right)^{2N_1+N_2} \left(\frac{1-\theta_1}{1-\theta_2}\right)^{2N_3+N_2} = 1$$

$$1 - \frac{\theta_1 - \theta_2}{1 - \theta_2}$$

$$\theta_1 - \theta_2 = 0$$

$$M_{\text{impr}} \neq 2N_1 + N_2$$

3)

$$P(\vec{X}|\theta) = \theta^{2(N_1^x - N_1^y)} (2\theta(1-\theta))^{N_2^x - N_2^y} (1-\theta)^{2(N_3^x - N_3^y)}$$

$$N_1^x = N_1^y$$

$$N_2^x = N_2^y$$

$$\text{ofc } N_3 = 100 - N_1 - N_2$$

$$N_3^x = N_3^y \rightarrow \text{M2 or Mcon}$$

$$T(X) = (N_1, N_2)$$

$$= \left( \sum_{i: X_i=1} 1, \sum_{i: X_i=2} 1 \right) = (49, 32)$$